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A Project Report on

**“AUTOMATIC FIRE FIGHTING SYSTEM IN
EV VECHICLES GIVE FOR THIS PROJECT”**

**AUTOMATIC FIRE FIGHTING SYSTEM IN EV
VECHICLES GIVE FOR THIS PROJECT**

Submitted in partial fulfilment for the award of the degree of

**BACHELOR OF TECHNOLOGY IN
COMPUTER SCIENCE & ENGINEERING- CSE**

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LIST OF ABBREVIATIONS

EV – Electric Vehicle
BMS – Battery Management System
LM35 – Temperature Sensor
MQ-2 – Smoke Sensor
DC – Direct Current
AC – Alternating Current
PCB – Printed Circuit Board

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ABSTRACT

Electric vehicles (EVs) are becoming increasingly popular due to their environmental benefits and energy efficiency. However, EV battery systems are highly sensitive to overheating, short circuits, and thermal runaway conditions, which can lead to dangerous fire accidents. Traditional fire safety methods in vehicles mainly depend on manual fire extinguishers and human intervention, which may not respond quickly during emergencies. Therefore, there is a growing need for an intelligent and automatic fire detection and suppression system in EV vehicles.

The proposed solution integrates temperature, smoke, and flame sensors with a microcontroller-based embedded system to continuously monitor the vehicle environment for fire-related conditions. The system detects abnormal temperature rise, smoke presence, and fire flames in real time. Once fire is detected, the controller automatically activates the fire suppression mechanism such as a water pump or extinguisher unit to control the fire immediately. Simultaneously, the system generates alerts through buzzer and notification modules to warn passengers and drivers about the emergency situation.

To improve system efficiency and reliability, the proposed model ensures continuous monitoring and rapid response with minimal human intervention. The hardware components are integrated using embedded programming techniques to provide accurate detection and automatic operation. The system also supports real-time monitoring through a user-friendly interface, allowing users to observe sensor readings and system status effectively.

The project aims to enhance passenger safety, reduce vehicle damage, and minimize fire-related risks in electric vehicles. Overall, the proposed Automatic Fire Fighting System in EV Vehicles provides a cost-effective, reliable, and intelligent safety solution for modern EV transportation systems.

Chapter 1

Introduction

Electric vehicles (EVs) are rapidly gaining popularity due to their environmental benefits, energy efficiency, and reduced dependence on fossil fuels. However, the increasing use of high-capacity lithium-ion batteries in EVs has introduced serious safety concerns related to battery overheating, short circuits, and thermal runaway conditions. These issues can result in sudden fire accidents, causing severe damage to vehicles, risk to passengers, and economic losses.

Traditional fire safety systems in vehicles mainly rely on manual fire extinguishers and human intervention, which may not respond effectively during emergency situations. In many cases, delays in detecting and controlling fire can lead to rapid fire spread and dangerous explosions. Therefore, there is a strong need for an intelligent and automatic fire fighting system that can detect fire hazards at an early stage and immediately activate suppression mechanisms to reduce damage and improve passenger safety.

To address this problem, this project presents an **Automatic Fire Fighting System in EV Vehicles** using embedded systems and sensor technology. The proposed system integrates temperature sensors, smoke sensors, and flame sensors to continuously monitor the vehicle environment for abnormal conditions. A microcontroller-based control unit processes the sensor data and automatically activates the fire suppression mechanism whenever fire or overheating is detected. The system also provides alert notifications through buzzer alarms and warning indicators to inform passengers and drivers during emergency situations.

The project adopts a real-time monitoring and automatic response approach to ensure rapid fire detection and suppression with minimal human intervention. The entire system is designed using embedded hardware components and control programming techniques, making it suitable for practical implementation in EV safety applications. The proposed solution aims to improve vehicle safety, minimize fire-related damage, and provide a reliable and cost-effective fire protection system for electric vehicles.

1.1 Background & Motivation

The rapid growth of the electric vehicle industry has increased the demand for advanced safety systems capable of handling battery-related emergencies. Lithium-ion batteries used in EVs are highly efficient but are also sensitive to overheating, overcharging, internal short circuits, and mechanical damage. These conditions may lead to thermal runaway, where excessive heat generation causes uncontrollable fire incidents. Such accidents pose serious threats to passenger safety and vehicle reliability.

Existing fire protection methods in vehicles are mostly manual and depend on the quick reaction of drivers or passengers. However, during emergencies, human response may

not always be fast enough to control the fire at its initial stage. Additionally, conventional systems lack continuous monitoring and automatic suppression capabilities, making them less effective for EV applications.

The motivation behind this project is to develop an intelligent, automatic, and reliable fire fighting system that can detect fire hazards in real time and respond immediately without human intervention. By integrating multiple sensors and automated suppression mechanisms, the system aims to provide enhanced protection for EV vehicles and passengers. The project also focuses on reducing fire-related risks, preventing severe vehicle damage, and improving overall transportation safety through smart embedded technology solutions.

1.2 Problem Statement

Electric vehicles are vulnerable to fire accidents due to battery overheating, electrical short circuits, and thermal runaway conditions. Existing vehicle safety systems mainly depend on manual fire extinguishers and human response, which may not be sufficient during emergencies. Delays in detecting and controlling fire can lead to severe vehicle damage, passenger injuries, and safety hazards. Therefore, there is a need for an intelligent and automatic fire fighting system that can detect fire-related conditions in real time and activate suppression mechanisms immediately to improve EV safety and reliability.

1.3 Objectives

1. To design and develop an automatic fire fighting system for EV vehicles using embedded systems and sensor technology.
 2. To detect abnormal temperature, smoke, and flame conditions in real time using multiple sensors.
 3. To automatically activate the fire suppression mechanism during emergency situations.
 4. To provide warning alerts and notifications to passengers and drivers for improved safety.
 5. To reduce fire-related risks, vehicle damage, and improve overall EV safety performance.
-

1.4 Scope of the Study

1. The project focuses on detecting fire hazards in EV vehicles using temperature, smoke, and flame sensors.

2. The system is designed to automatically activate fire suppression mechanisms when fire conditions are detected.
 3. The project includes real-time monitoring and alert systems for emergency notifications.
 4. The proposed system aims to provide a low-cost, reliable, and efficient fire protection solution for electric vehicles.
-

1.5 Major Contributions

1. Development of a real-time fire detection system using multiple sensors for EV safety applications.
2. Integration of automatic fire suppression mechanisms for rapid emergency response.
3. Implementation of embedded control systems for intelligent monitoring and operation.
4. Enhancement of passenger safety through automated alerts and emergency handling.
5. Design of a practical and cost-effective fire fighting solution suitable for modern electric vehicles.

CHAPTER 2

LITERATURE SURVEY 2.1

The rapid development of electric vehicles (EVs) has brought significant improvements in sustainable transportation and energy efficiency. However, the widespread use of lithium-ion batteries in EVs has introduced new safety challenges, particularly in relation to fire hazards. Lithium-ion batteries have high energy density and are capable of storing large amounts of electrical energy within a compact structure. While this property makes them ideal for electric vehicles, it also increases the risk of overheating, short circuits, and thermal runaway conditions.

Thermal runaway is one of the most critical safety concerns in lithium-ion battery systems. It occurs when a battery cell experiences uncontrolled temperature rise due to internal chemical reactions. Once thermal runaway begins, the heat generated can spread rapidly to neighboring battery cells, leading to a chain reaction that can result in fire or explosion. Several EV fire incidents reported globally have emphasized the need for effective fire detection and suppression systems within electric vehicles.

Leading EV manufacturers such as Tesla, Inc. have implemented advanced battery management systems to monitor battery temperature, voltage, and current levels. These systems are designed to detect abnormal operating conditions and isolate faulty battery modules. However, many existing systems mainly focus on prevention rather than active fire suppression. Once a fire begins inside the battery pack, controlling it becomes extremely difficult without specialized suppression systems.

Research studies have shown that early detection of fire indicators such as temperature increase, smoke formation, and gas emissions can significantly reduce the risk of catastrophic failure. Sensors capable of detecting these parameters can be integrated with embedded systems to provide real-time monitoring of battery compartments. When abnormal conditions are detected, the system can activate protective mechanisms such as alarms, cooling systems, or fire suppression devices.

Another important aspect highlighted in the literature is the role of Battery Management Systems (BMS). The BMS is responsible for monitoring battery health and preventing overcharging, over-discharging, and short circuits. Although BMS technology has improved significantly in recent years, most systems still lack an integrated automatic fire suppression mechanism. Therefore, there is a growing interest in developing intelligent fire detection and suppression systems that can work alongside the BMS.

Several researchers have also explored the use of smoke sensors and gas sensors to detect early stages of battery failure. When lithium-ion batteries begin to degrade or overheat, they release gases such as carbon monoxide and other volatile compounds. Gas detection sensors can identify these emissions and provide early warning signals

before a fire develops. This approach improves the reliability of fire detection systems by identifying hazardous conditions at an earlier stage.

In addition to gas detection, flame sensors and infrared sensors have been studied for their ability to detect open flames. These sensors are capable of identifying specific wavelengths of light emitted by flames and can quickly trigger fire suppression mechanisms. By combining multiple sensors such as temperature sensors, smoke sensors, and flame sensors, researchers have developed multi-parameter monitoring systems that improve detection accuracy and reduce false alarms.

Automotive safety companies such as Bosch and other technology providers have developed advanced monitoring solutions that integrate sensor networks with embedded controllers. These systems continuously monitor battery temperature, current flow, and environmental conditions inside battery enclosures. However, most commercial systems are designed for high-end vehicles and are relatively expensive, which limits their adoption in low-cost electric vehicles.

Another area of research focuses on fire suppression technologies suitable for electric vehicles. Traditional fire extinguishers use water or foam, which may not be ideal for lithium-ion battery fires. Some researchers have investigated the use of carbon dioxide (CO₂) and dry chemical agents that can effectively suppress flames without causing electrical damage. Water mist systems have also been studied because they can rapidly cool battery surfaces and prevent thermal propagation between cells.

Despite these advancements, many challenges remain in implementing effective fire safety solutions in EVs. One major challenge is the compact design of battery packs, which limits the space available for installing fire suppression equipment. Additionally, battery compartments are often sealed to protect them from environmental factors, making it difficult to detect and suppress fires internally.

Another challenge is the cost and complexity associated with advanced fire safety systems. Industrial-grade suppression systems used in large battery storage facilities are often expensive and require specialized installation. For small-scale EV applications, researchers are exploring low-cost embedded systems that use microcontrollers and sensors to provide automated fire detection and suppression.

Recent studies suggest that integrating multi-sensor detection systems with microcontroller-based control units can provide an effective and economical solution for EV fire safety. These systems monitor several parameters simultaneously and activate suppression mechanisms when predefined threshold conditions are exceeded. The use of embedded controllers such as Arduino or other microcontroller platforms allows rapid processing of sensor data and immediate response to hazardous conditions.

Therefore, the development of a compact, automated, and cost-effective fire fighting system for electric vehicles is an important research area. By combining temperature

sensors, smoke detectors, and flame sensors with a microcontroller-based control system, it is possible to detect early signs of fire and activate suppression mechanisms automatically. Such systems can significantly improve the safety of electric vehicles and reduce the risk of battery fire incidents

Table 2.1 Comparison of Existing Fire Detection Systems

System Type	Detection Method	Response Time	Limitations
Manual Fire Extinguisher	Human detection	Slow	Requires human intervention
Smoke Detector	Smoke sensing	Moderate	Cannot detect temperature rise early
Temperature Monitoring	Thermal sensors	Moderate	Does not detect smoke or flame
Gas Detection Systems	Gas emission sensing	Moderate	Requires calibration
Multi-Sensor Systems	Temperature + Smoke + Flame	Fast	Slightly higher complexity

Table 2.2 Comparison of Existing and Proposed Fire Protection Systems

Parameter	Existing Systems	Proposed System
Fire Detection	Limited sensors	Multiple sensors
Automation	Partial	Fully automatic
Response Time	Slow	Faster response
Cost	High	Low-cost prototype
Integration	Separate systems	Integrated system

Table 2.3 Summary of Research Studies on EV Fire Safety

Research Area	Focus	Outcome
Thermal Monitoring	Battery temperature control	Prevent overheating
Gas Detection	Early fault detection	Early warning signals

Research Area	Focus	Outcome
Flame Detection	Infrared sensing	Quick flame identification
Fire Suppression	CO ₂ / Water mist	Flame control and cooling

CHAPTER 3

EXISTING SYTEM

The safety of electric vehicles has become an important concern as the adoption of EV technology continues to increase worldwide. Lithium-ion batteries, which are commonly used in EVs, are known for their high energy density and long life cycle. However, they are also sensitive to temperature variations, electrical faults, and mechanical damage. If these batteries are exposed to extreme conditions, they may lead to overheating, fire hazards, or even explosions.

Existing fire protection systems in electric vehicles mainly focus on monitoring battery conditions and preventing abnormal operation. These systems generally rely on Battery Management Systems (BMS) that monitor parameters such as voltage, current, and temperature. The BMS continuously checks whether the battery is operating within safe limits and attempts to prevent faults by disconnecting the battery when abnormal conditions occur.

Although Battery Management Systems are effective in preventing certain types of battery failures, they do not provide complete fire protection. In many cases, the system only alerts the driver or shuts down the battery circuit. Once a fire has started inside the battery pack, it becomes very difficult to control it because the battery modules are tightly packed inside a sealed enclosure.

Many electric vehicles currently use passive fire protection techniques. Passive safety systems are designed to reduce the risk of fire by improving battery insulation, providing thermal barriers between battery cells, and controlling battery temperature through cooling systems. These methods help delay the spread of heat but do not actively extinguish fires once they occur.

Some modern electric vehicles also use thermal monitoring sensors placed inside the battery pack. These sensors measure the temperature of battery cells and send data to the vehicle control unit. If the temperature exceeds a safe threshold, the system can trigger protective actions such as shutting down the battery system or activating cooling fans. While this approach helps in preventing overheating, it cannot suppress flames or smoke once ignition occurs.

Another type of fire detection system used in industrial battery storage facilities involves smoke detection and gas detection sensors. These sensors detect gases released during battery failure or thermal runaway conditions. However, these systems are usually expensive and complex, making them unsuitable for small electric vehicles or low-cost transportation systems.

Automotive technology providers such as Bosch and Continental AG have developed advanced battery monitoring technologies for electric vehicles. These systems focus on

battery diagnostics and predictive maintenance. While they provide accurate monitoring and fault detection, they do not always include automatic fire suppression mechanisms. Furthermore, traditional fire extinguishing methods used in internal combustion engine vehicles are not always effective for lithium-ion battery fires. Water-based extinguishers can sometimes cause electrical hazards, while foam extinguishers may not penetrate sealed battery packs effectively. This creates a need for specialized fire suppression techniques designed specifically for EV battery compartments.

Another limitation of existing systems is their dependency on human intervention. In many cases, the driver must manually respond to warning alerts or use a fire extinguisher when a fire occurs. However, during emergencies, the driver may not be able to react quickly enough to prevent damage.

Due to these limitations, there is a growing demand for automated fire detection and suppression systems that can operate without human intervention. Such systems should be capable of detecting early signs of fire, triggering alarms, and activating suppression mechanisms automatically.

3.1 Working of Existing Fire Protection Systems

The existing fire protection systems in electric vehicles primarily operate through monitoring and preventive mechanisms. These systems are integrated with the vehicle's Battery Management System and rely on sensors to measure battery parameters.

The working process generally follows the steps below:

1. Temperature sensors continuously monitor battery temperature.
2. Voltage sensors monitor individual battery cell voltages.
3. Current sensors measure the flow of electrical current.
4. The Battery Management System analyzes the collected data.
5. If abnormal conditions are detected, the system disconnects the battery or alerts the driver.

Although this process helps prevent certain types of failures, it does not actively suppress fires.

3.2 Limitations of Existing Systems

Despite advancements in battery safety technology, several limitations exist in current fire protection systems.

First, most systems rely only on monitoring parameters rather than directly detecting fire or smoke. This means that some fire hazards may not be detected until it is too late.

Second, the response time of manual fire suppression methods is relatively slow. In emergency situations, delays in response can allow the fire to spread rapidly within the battery pack.

Third, existing systems often require human intervention, which may not always be possible during accidents or sudden battery failures.

Fourth, many advanced fire suppression technologies are expensive and difficult to integrate into compact electric vehicles.

Finally, current systems do not provide integrated solutions that combine fire detection, alarm generation, and automatic fire suppression.

Table 3.1 Existing EV Fire Safety Methods

Table 3.1 Existing EV Fire Safety Methods			
Method	Description	Advantages	Limitations
Battery Management System	Monitors voltage, temperature and current	Prevents battery damage	Does not suppress fire
Thermal Monitoring	Detects high temperature	Early warning	Limited fire detection
Smoke Detection	Detects smoke particles	Simple detection	Cannot detect early heat
Manual Fire Extinguishers	Used by driver	Simple to operate	Requires human action

Table 3.2 Problems in Existing Fire Protection Systems

Problem	Impact
Lack of automatic fire suppression	Fire spreads quickly
Delayed detection	Late response to hazards
Dependence on manual action	Human error possible
High system cost	Not suitable for low-cost EVs

CHAPTER 4

PROPOSED SYSTEM

The proposed system aims to provide an automated solution for detecting and suppressing fire hazards in electric vehicles. Unlike conventional fire safety mechanisms that rely on manual intervention, the proposed system is designed to operate automatically by continuously monitoring the environmental conditions inside the electric vehicle battery compartment.

The system uses a combination of sensors and an embedded control unit to detect early signs of fire such as temperature rise, smoke formation, and flame presence. These sensors are connected to a microcontroller that processes the sensor data in real time. If any abnormal condition is detected, the system immediately activates a fire suppression mechanism and alerts the user through an alarm.

The proposed system is designed to be compact, cost-effective, and suitable for integration with existing electric vehicle safety systems. By implementing multiple sensors, the reliability and accuracy of fire detection are improved. The system can detect fire hazards at an early stage and take preventive actions before the situation becomes critical.

4.1 Introduction to the Proposed Model

The Automatic Fire Fighting System for Electric Vehicles is designed to detect fire hazards and suppress them automatically. The system integrates temperature sensors, smoke sensors, and flame sensors with a microcontroller-based control unit. These sensors continuously monitor the battery environment and send data to the controller.

When the sensor readings exceed predefined threshold values, the microcontroller activates a relay that controls a water pump or fire suppression unit. At the same time, an alarm is triggered to notify the driver about the fire hazard.

This automated approach reduces human dependency and ensures a faster response to fire emergencies.

4.2 System Architecture

The architecture of the proposed system consists of three major modules:

1. Detection Module
2. Control Module
3. Suppression Module

Detection Module

The detection module is responsible for identifying fire-related parameters. It consists of the following sensors:

- Temperature Sensor (LM35) – Detects abnormal temperature rise.
- Smoke Sensor (MQ-2) – Detects smoke and flammable gases.
- Flame Sensor – Detects infrared radiation emitted by flames.

These sensors are placed near the battery compartment where the risk of fire is highest.

Control Module

The control module is the central processing unit of the system. It consists of an Arduino Uno microcontroller that receives signals from the sensors and processes them. The microcontroller compares the sensor readings with predefined safety thresholds. If the sensor values exceed the safe limit, the controller sends a signal to activate the suppression module.

Suppression Module

The suppression module is responsible for controlling the fire. It consists of:

- Relay module
- Water pump or fire suppression unit
- Buzzer alarm

When activated, the relay switches on the pump to spray water or fire suppression liquid on the battery area.

4.3 Block Diagram of the Proposed System

The block diagram represents the overall structure of the proposed system and shows how different components interact with each other.

Working flow:

Sensors → Microcontroller → Relay → Pump & Alarm

The sensors continuously send signals to the microcontroller. The microcontroller processes these signals and determines whether the conditions are safe or hazardous. If a fire hazard is detected, the microcontroller activates the relay, which in turn starts the pump and alarm system.

4.4 Working Principle

The working principle of the proposed fire fighting system is based on real-time monitoring and automatic response.

The system begins operation when power is supplied to the microcontroller and sensors. The sensors continuously monitor the temperature, smoke levels, and flame presence in the battery compartment. The sensor outputs are sent to the Arduino microcontroller, which processes the data.

If the temperature rises beyond the threshold level or smoke is detected, the microcontroller immediately interprets this as a potential fire hazard. When such conditions occur, the controller activates the relay module. The relay switches on the water pump or fire suppression system, which sprays extinguishing material over the affected area.

At the same time, a buzzer alarm is activated to alert the driver and passengers about the emergency. The system continues operating until the sensor readings return to safe levels.

This automatic mechanism ensures a quick response to fire hazards and helps prevent major damage.

4.5 Advantages of the Proposed System

The proposed system offers several advantages compared to existing fire protection methods.

First, the system provides automatic fire detection and suppression, eliminating the need for human intervention. This significantly reduces response time during emergencies.

Second, the use of multiple sensors improves detection accuracy and minimizes false alarms. By monitoring temperature, smoke, and flame simultaneously, the system can identify fire hazards more reliably.

Third, the system is cost-effective and easy to implement, making it suitable for both small electric vehicles and larger EV systems.

Fourth, the system can be integrated with existing battery management systems for improved safety.

Finally, the proposed system improves overall vehicle safety by preventing fire accidents and reducing damage caused by battery failures.

Table 4.1 Components of the Proposed System

Component	Function
LM35 Sensor	Detects temperature changes
MQ-2 Sensor	Detects smoke and gas

Component	Function
Flame Sensor	Detects fire flames
Arduino Uno	Processes sensor data
Relay Module	Controls pump operation
Water Pump	Sprays fire suppression liquid
Buzzer	Provides alarm notification

Table 4.2 Advantages of Proposed System

Feature	Existing System	Proposed System
Fire Detection	Limited monitoring	Multi-sensor detection
Automation	Partial	Fully automatic
Response Time	Slow	Fast
Safety Level	Moderate	High

CHAPTER 5

IMPLEMENTATION

The implementation phase of the Automatic Fire Fighting System for Electric Vehicles involves designing and integrating hardware components that can detect fire hazards and respond automatically. The system is implemented using sensors, a microcontroller, and a fire suppression mechanism. These components work together to detect abnormal conditions such as high temperature, smoke, or flame inside the electric vehicle battery compartment.

The main objective of the implementation is to develop a working prototype that can monitor environmental conditions and activate fire suppression when necessary. The system is designed in such a way that it can continuously monitor the battery environment and respond quickly in case of any fire-related hazards.

The implementation process includes hardware setup, sensor integration, microcontroller programming, and system testing.

5.1 Hardware Components

The hardware components form the foundation of the proposed system. Each component performs a specific function in detecting and controlling fire hazards.

Arduino Uno

Arduino Uno is the main controller used in this project. It is a microcontroller board based on the ATmega328P microcontroller. The Arduino receives signals from sensors and processes them according to the programmed instructions.

The Arduino is responsible for analyzing the sensor readings and determining whether the system should activate the fire suppression mechanism. It also controls the relay module and alarm system.

Temperature Sensor (LM35)

The LM35 temperature sensor is used to measure the temperature inside the electric vehicle battery compartment. This sensor provides an analog output voltage that is proportional to the temperature.

If the temperature rises beyond the predefined threshold, it indicates a possible overheating condition. The sensor sends this information to the Arduino controller, which then triggers the fire protection system if necessary.

Smoke Sensor (MQ-2)

The MQ-2 sensor is used to detect smoke and flammable gases that may be released during battery failure. Lithium-ion batteries often release gases before catching fire. Therefore, detecting these gases can help in identifying fire hazards at an early stage. The MQ-2 sensor continuously monitors the air quality around the battery compartment and sends signals to the microcontroller when smoke is detected.

Flame Sensor

The flame sensor detects infrared radiation emitted by flames. It is used to identify the presence of open flames inside the vehicle compartment. This sensor improves the reliability of the fire detection system by providing an additional layer of safety. If a flame is detected, the sensor immediately sends a signal to the Arduino controller.

Relay Module

The relay module acts as a switch that allows the microcontroller to control high-power devices such as pumps or motors. When the Arduino detects a fire hazard, it activates the relay. The relay then switches on the fire suppression system.

Water Pump / Fire Suppression Unit

The water pump or suppression unit is responsible for extinguishing the fire. When activated, the pump sprays water or fire suppression liquid over the affected area. This helps reduce the temperature and suppress the flames before the fire spreads further.

Buzzer Alarm

The buzzer is used as an alert system. When a fire hazard is detected, the buzzer produces a loud sound to notify the driver and passengers about the emergency. This warning allows people inside the vehicle to take immediate safety actions.

Table 5.1 Hardware Components Used

Component	Purpose
Arduino Uno	Main control unit
LM35 Temperature Sensor	Detects temperature rise

Component	Purpose
MQ-2 Smoke Sensor	Detects smoke and gas
Flame Sensor	Detects fire flames
Relay Module	Controls pump activation
Water Pump	Suppresses fire
Buzzer	Provides alarm

5.2 Circuit Design

The circuit design connects all sensors and components to the Arduino microcontroller. Each sensor is connected to specific input pins of the Arduino board.

The temperature sensor sends analog signals to the Arduino, while the smoke sensor and flame sensor provide digital signals. The Arduino processes these signals and determines whether the system should activate the fire suppression mechanism.

The relay module is connected to one of the digital output pins of the Arduino. When the relay receives a signal from the Arduino, it switches on the water pump and buzzer.

The power supply for the system can be provided through a battery or external power source.

5.3 Software Implementation

The software part of the project is developed using the Arduino Integrated Development Environment (IDE). The program is written in the Arduino programming language, which is based on C/C++.

The Arduino program performs the following functions:

1. Initializes the sensors and input/output pins.
2. Continuously reads data from temperature, smoke, and flame sensors.
3. Compares sensor readings with predefined threshold values.
4. Determines whether a fire hazard exists.
5. Activates the relay and buzzer when danger is detected.

The program runs continuously in a loop so that the system can monitor conditions in real time.

5.4 System Testing

After assembling the hardware and uploading the program to the Arduino, the system is tested to verify its functionality.

During testing, different conditions are simulated to observe how the system responds. For example, heat sources can be used to increase temperature near the sensor, or smoke can be introduced to test the smoke detection capability.

When the sensors detect abnormal conditions, the system should automatically activate the relay and start the water pump. The buzzer should also produce an alarm sound to alert users.

Successful testing confirms that the system works correctly and can respond to fire hazards effectively.

Table 5.2 Testing Scenarios

Test Condition	Expected Response
High Temperature	System activates pump
Smoke Detection	Alarm and pump activated
Flame Detection	Immediate fire suppression
Normal Condition	System remains idle

CHAPTER 6

CONCLUSION & FUTURE ENHANCEMENTS

The Automatic Fire Fighting System for Electric Vehicles was successfully designed and implemented using sensors, a microcontroller, and a fire suppression mechanism. The system was tested under different environmental conditions to evaluate its performance and reliability. The primary goal of the testing process was to verify whether the system could accurately detect fire hazards and respond automatically.

During testing, the temperature sensor, smoke sensor, and flame sensor were used to simulate different fire-related scenarios. The sensors continuously monitored the surrounding environment and transmitted data to the Arduino microcontroller. The controller processed the sensor signals and compared them with predefined threshold values.

When the temperature was increased beyond the safe limit, the LM35 temperature sensor detected the abnormal rise in temperature. The sensor sent a signal to the Arduino controller, which then activated the relay module. As a result, the water pump was switched on and the buzzer alarm was activated to indicate a fire hazard.

Similarly, when smoke was introduced near the MQ-2 smoke sensor, the sensor detected the presence of smoke particles and sent a signal to the microcontroller. The system responded immediately by activating the alarm and fire suppression mechanism. This demonstrated the effectiveness of the smoke detection module in identifying potential fire hazards.

The flame sensor was also tested by exposing it to a small flame source. The sensor successfully detected the infrared radiation emitted by the flame and triggered the fire suppression system instantly. This confirmed that the flame detection module was functioning correctly.

The testing results indicate that the system is capable of detecting fire hazards at an early stage and responding quickly. The combination of multiple sensors improves the reliability of the system because it monitors different fire indicators simultaneously.

Another important observation during testing was the response time of the system. The microcontroller processed the sensor data almost instantly, allowing the relay and pump to activate within a very short period. This quick response is essential in preventing the spread of fire inside electric vehicle battery compartments.

The results show that the proposed system can significantly enhance the safety of electric vehicles by providing an automatic fire detection and suppression mechanism. The prototype demonstrated reliable performance and was able to operate effectively under different test conditions.

Table 6.1 Experimental Results

Test Scenario	Sensor Triggered	System Response	Result
High Temperature	LM35 Sensor	Pump and alarm activated	Successful
Smoke Presence	MQ-2 Sensor	Alarm and pump activated	Successful
Flame Detection	Flame Sensor	Immediate suppression	Successful
Normal Conditions	None	System idle	Successful

Table 6.2 System Performance Analysis

Parameter	Performance
Detection Accuracy	High
Response Time	Very Fast
System Reliability	Good
Fire Suppression Efficiency	Effective

CHAPTER-7

CONCLUSION AND FUTURE SCOPE

Conclusion

The Automatic Fire Fighting System for Electric Vehicles was successfully designed and implemented to enhance the safety of electric vehicle battery systems. The system integrates multiple sensors, a microcontroller, and a fire suppression mechanism to detect and control fire hazards automatically.

Electric vehicles rely heavily on lithium-ion batteries, which are known for their high energy density. However, these batteries are also prone to overheating and thermal runaway conditions that may lead to fire hazards. Therefore, developing an efficient fire safety system is essential for ensuring the safe operation of electric vehicles.

The proposed system uses temperature sensors, smoke sensors, and flame sensors to continuously monitor the battery environment. The sensors send real-time data to the Arduino microcontroller, which processes the information and identifies potential fire hazards. When abnormal conditions are detected, the system activates a relay module that controls the fire suppression unit.

One of the major advantages of this system is its ability to operate automatically without human intervention. This ensures faster response during emergency situations and reduces the risk of fire spreading within the vehicle. The buzzer alarm also provides an immediate warning to the driver and passengers.

The system was tested under different conditions such as high temperature, smoke presence, and flame detection. The experimental results showed that the system was able to detect fire hazards accurately and activate the suppression mechanism within a short time. This demonstrates that the proposed system can effectively improve fire safety in electric vehicles.

Overall, the developed prototype proves that integrating sensor-based monitoring with automated fire suppression can significantly enhance vehicle safety. The system is simple, cost-effective, and suitable for implementation in electric vehicles.

Future Scope

Although the proposed system performs effectively, there are several improvements that can be made in the future to enhance its functionality and reliability.

One possible improvement is the integration of advanced sensors capable of detecting battery gas emissions more accurately. These sensors can identify early signs of battery failure before a fire occurs.

Another improvement is the addition of wireless communication technology such as IoT modules. By integrating an Internet of Things (IoT) platform, the system can send real-time alerts to the vehicle owner or monitoring center through a mobile application.

The system can also be enhanced by integrating it with the vehicle's Battery Management System. This integration will allow the system to monitor battery parameters more precisely and provide better protection.

In the future, more advanced fire suppression technologies such as automatic aerosol fire extinguishing systems can be used instead of water pumps. These systems are more effective in controlling lithium-ion battery fires.

Artificial intelligence and machine learning techniques can also be implemented to analyze sensor data and predict potential fire hazards before they occur.

Therefore, further research and development in this field can lead to more advanced and reliable fire protection systems for electric vehicles.

REFERENCES

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 2. A. Emadi, *Advanced Electric Drive Vehicles*, CRC Press.
 3. Research papers on lithium-ion battery safety and fire suppression systems.
 4. Arduino Official Documentation – Sensor Interfacing.
 5. Studies on thermal runaway in lithium-ion batteries.
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APPENDIX

Sample Arduino Program (Example)

```
int tempSensor = A0;  
int smokeSensor = A1;  
int flameSensor = 7;  
int relay = 8;  
int buzzer = 9;
```

```
void setup()
{
  pinMode(flameSensor, INPUT);
  pinMode(relay, OUTPUT);
  pinMode(buzzer, OUTPUT);
  Serial.begin(9600);
}

void loop()
{
  int tempValue = analogRead(tempSensor);
  int smokeValue = analogRead(smokeSensor);
  int flameValue = digitalRead(flameSensor);

  if(tempValue > 500 || smokeValue > 400 || flameValue == LOW)
  {
    digitalWrite(relay, HIGH);
    digitalWrite(buzzer, HIGH);
  }
  else
  {
    digitalWrite(relay, LOW);
    digitalWrite(buzzer, LOW);
  }

  delay(500);
}
```